Sarah Oberbichler, Eva Pfanzelter

# Tracing Discourses in Digital Newspaper Collections

## A Contribution to Digital Hermeneutics while Investigating 'Return Migration' in Historical Press Coverage

**Abstract:** Based on a case study on return migration – defined as the movement of emigrants, refugees, prisoners of war, and others back to their country of origin – this paper deals with a historical-critical approach to digital newspaper collections and interfaces. Using discourse driven historical research questions to examine return migration, it highlights the usability and relevance of digital source criticism (OCR issues, digitization process, newspaper title selection, etc.), digital query criticism (selection of keywords or parts of keywords, consequences of length, polysemy, etc.), and interface criticism (possibilities to search, view, select, collect, visualize, contextualization-efforts, metadata, etc.). We argue that due to the complexity of language coupled with intransparent functionalities of newspaper interfaces, distorted, misunderstood or misinterpreted results can occur. Some inherent biases in the digital collections themselves as well as missing metadata and context can lead to distortions that only at a first glance seem trivial and banal.

**Keywords:** digitized newspapers, return migration, digital hermeneutics, digital history, newspaper user interfaces

## 1 Introduction

The digitisation of historical newspapers and their access through online interfaces provided by national or regional libraries offers many opportunities for different user groups: researchers, students, teachers, lay historians, and other interested people can use online interfaces to research, for example, specific topics, their family or community history, events in the press or changes of interpretations over time. Institutions such as the Austrian National Library (ONB), the Bibliothèque nationale de France (BnF) or the National Library of Finland, to name only a

**Acknowledgment:** This work has been supported by the European Union's Horizon 2020 research and innovation program under grant 770299 (NewsEye). We thank our colleague Stefan Hechl from the NewsEye project for proofreading.

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https://doi.org/10.1515/9783110729214-007

few, have engaged and invested heavily in newspaper digitisation. With the Europeana Newspaper's collection,[1] there is also a pan-European newspaper portal available that includes digitized newspapers from 20 countries, dating from 1618 to 1996. Practical considerations such as the availability of thousands of issues of newspapers in libraries as well as non-existent copyright restrictions for material older than 100 years were decisive factors for digitizing newspaper collections. Digitization was also fostered by the appreciation of the potential of newspapers as rich sources for research on history, society and language. For libraries, making newspapers digitally available had and still has several implications. On the one hand, they reach out to a potentially global audience, and on the other they can (usually) preserve the fragile original printed versions in their archives or reserve access to limited in-house users.[2]

Before digitisation, working with newspapers oftentimes was a daunting task due to the amount of material and the time that had to be invested in order to find information that goes beyond reactions to a specific event or a known date. It included spending long hours at library desks and going through large numbers of documents without inventories or any kind of structure.[3] With the advances in digitisation and text recognition, digital approaches have gained in relevance for humanities research and the social sciences, linguistics or history, while historical corpora have become an interesting field of investigation for computer scientists. Interdisciplinary projects which bring together cultural heritage institutions with researchers from the humanities and from computer science are multiplying. For example, the importance of context and meaning is slowly gaining ground in big data research and use. Equally, a critical approach to digital searching and text mining enables researchers to explore and interpret large amounts of newspaper data in novel ways, to visualize quantitatively and to perform individual qualitative analysis.[4] As a result, newly developed methods and tools promise to change the way digital heritage data is accessed, (re)searched, used and analysed.

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**1** https://www.europeana.eu/de/collections/topic/18-newspapers.

**2** Natasha Stroeker and René Vogels (2012). *Survey Report on Digitisation in European Cultural Heritage Institutions 2012*. Tech. rep., p. 25. URL: http://enumeratedataplatform. digibis.com/reports/core-survey-i-final-report/detail; Clemens Neudecker and Apostolos Antonacopoulos (2016). "Making Europe's Historical Newspapers Searchable." In: *2016 12th IAPR Workshop on Document Analysis Systems (DAS)*. Santorini, Greece: IEEE, pp. 405–410. DOI: 10.1109/DAS.2016.83; Eleni Galiotou (2014). "Using digital corpora for preserving and processing cultural heritage texts: a case study." In: *Library Review* 63.6/7, pp. 408–421. DOI: 10.1108/LR-11-2013-0142.

**3** Richard Abel (2013). "The Pleasures and Perils of Big Data in Digitized Newspapers." In: *Film History* 25.1, pp. 1–10. DOI: 10.2979/filmhistory.25.1–2.1.

**4** Adrian Bingham (2010). "The Digitization of Newspaper Archives: Opportunities and Challenges for Historians." In: *Twentieth Century British History* 21.2, pp. 225–231.

Advances in digitization also call for an update of the historical-critical method, the hermeneutical process which guides the investigation of the past through three main steps: heuristics, source criticism and interpretation.[5] With regards to the second step, the demand for digital source criticism (and it often goes hand in hand with tool criticism) by now is of long standing.[6] Other parts of this historical method, like the relevance and critical assessment of adequate corpus creation in the heuristic research step, have so far not been considered enough. For the third methodological phase, the interpretation of digital sources and corpora, we only begin to understand the possibilities offered by computer science and the far reaching consequences automation has on both the humanities' disciplines and on society. To use digitized newspaper corpora to dig deeper into the historical-critical method in the digital age is therefore a logical consequence.

In order to understand social and historical developments based on digital newspaper corpora, however, critical readers and interpreters are needed.[7] Many users are not aware of the biases that come along with the processing and datafication of historical newspapers as well as their access via interfaces.[8] Interfaces steer what users can learn from digitised newspapers (e.g., selective curation[9]) and they influence workflows by offering specific, pre-selected functions and tools. The

**5** Peter Leyh (1977). *Johann Gustav Droysen: Historik. Bd. 1: Rekonstruktion der ersten vollständigen Fassung der Vorlesungen (1857). Grundriß der Historik in der ersten handschriftlichen (1857/58) und in der letzten gedruckten Fassung (1882)*. Stuttgart-Bad Cannstatt. 532 pp.

**6** Andreas Fickers (2020). "Update für die Hermeneutik. Geschichtswissenschaft auf dem Weg zur digitalen Forensik?" In: *Zeithistorische Forschungen/Studies in Contemporary History* 17.1, pp. 157–168. DOI: 10.14765/ZZF.DOK-1765; Marijn Koolen, Jasmijn van Gorp, and Jacco van Ossenbruggen (2019). "Toward a model for digital tool criticism: Reflection as integrative practice." In: *Digital Scholarship in the Humanities* 34.2, pp. 368–385. DOI: 10. 1093/llc/fqy048; Eva Pfanzelter (2010). "Von der Quellenkritik zum kritischen Umgang mit digitalen Ressourcen." In: *Digitale Arbeitstechniken für Geistes- und Kulturwissenschaften*. Ed. by Martin Gasteiner and Peter Haber. UTB M (Medium Format) 3157. Stuttgart, pp. 39–50; Pascal Föhr (2017). "Historische Quellenkritik im Digitalen Zeitalter." Thesis. University of Basel. DOI: 10.5451/unibas-006805169.

**7** James Mussell (2012). *The Nineteenth-Century Press in the Digital Age*. London: Palgrave Macmillan UK.

**8** Elizabeth Toon (2019). "The tool and the job: Digital humanities methods and the future of the history of the human sciences." In: *History of the Human Sciences* 32.1, pp. 83–98. DOI: 10.1177/0952695119834152; Andrew Hobbs (2013). "The Deleterious Dominance of *The Times* in Nineteenth-Century Scholarship." In: *Journal of Victorian Culture* 18.4, pp. 472–497. DOI: 10.1080/13555502.2013.854519.

**9** Jon Coburn (2020). "Defending the digital: Awareness of digital selectivity in historical research practice." In: *Journal of Librarianship and Information Science*, pp. 1–14. DOI: 10.1177/0961000620918647.

possibilities to perform source criticism in this sense is affected by interfaces that select, capture, sort, and sometimes gather additional (outside) information.[10] Some studies indicate that interfaces do not always concur with the needs of different user groups. Users comment on missing metadata (e.g., digitization strategy, metrics of newspapers and issues, publication frequency, geographical area covered), request contextualization of the source material (e.g., additional information on newspapers, on political orientation, explanation of the search-results, historical context of both newspapers and results, etc.) and ask for references to the limitations of collections (e.g., missing issues, absent time-periods, bad paper quality and subsequently bad Optical Character Recognition [OCR]). Additionally, in these studies a demand for adequate tools to deal with newspapers and search results as well as for material for educational use was clearly identified.[11]

As the role of digital tools when working with newspapers grows, it is important for scholars to understand the biases inherent in digitization processes, digital interfaces and search strategies. While these are sometimes acknowledged as an issue, they are rarely discussed by means of concrete examples. Bearing these findings in mind, the goal of this chapter is to use one case study – the topic of the return of emigrants to their place of origin between 1850 and 1950 – to show both opportunities and limitations that come when working with digital newspapers via interfaces. It is argued that, in addition to still highly relevant issues about the quality of the source material and text recognition, the lack of transparency regarding both digital newspaper collections and interfaces introduces several biases. We also show how newspaper interfaces (e.g., contextualization, explanation and documentation) could effectively counter misunderstandings and misinterpretation. In order to underline our arguments we introduce some advanced text mining functionalities as methods of analysis and interpretation that have the potential to support academic or expert approaches, for example, when creating specific

**10** Johan Jarlbrink and Pelle Snickars (2017). "Cultural heritage as digital noise: nineteenth century newspapers in the digital archive." In: *Journal of Documentation* 73.6, pp. 1228–1243. DOI: 10.1108/JD- 09- 2016- 0106; Ludmilla Jordanova (2014). "Historical Vision in a Digital Age." In: *Cultural and Social History* 11.3, pp. 343–348. DOI: 10 . 2752 /147800414X13983595303237; Arn Keeling and John Sandlos (2011). "Shooting the Archives: Document Digitization for Historical-Geographical Collaboration: Shooting the Archives." In: *History Compass* 9.5, pp. 423–432. DOI: 10.1111/j.1478–0542.2011.00771.x.

**11** Maud Ehrmann, Estelle Bunout, and Marten Düring (2019). "Historical Newspaper User Interfaces: A Review." In: URL: https://zenodo.org/record/3404155; Sarah Oberbichler et al. (2019). *Online research of digital newspapers of three national libraries: A survey*. URL: https://www.newseye.eu/blog/news/online-research-of-digital-newspapers-of-threenational-libraries-a-survey-by-sarah-oberbichler-stef/.

corpora for in-depth research or for the classification of search results. This means we try to give answers to the question of how distortions and one-sided interpretations can be reduced already during the heuristic process. Thus the chapter underlines the importance of digital source, query and interface criticism by pointing out the biases that can arise when dealing with large quantities of digital data and it therefore is intended as a contribution to discussions about digital hermeneutics.

In this paper, the Austrian newspaper collections accessible trough the platform AustriaN Newspapers Online (ANNO)[12] was used to investigate the topic of return migration in the second half of the 19th century and first half of the 20th century. ANNO is the user interface for digital newspaper corpora run by the Austrian National Library. The platform offers free access to 21 million Austrian newspaper and magazine pages that were published between 1568 and 1950.

# 2 Discourses on Return Migration

## 2.1 Return Migration as a Topic of Historical Research

The definition of remigration or return migration is not consistent within research literature. However, all of the explanations of the term have in common that they define return migration as 'cross-border migration to the country of origin',[13] or as the return of a person to the country of origin after spending a longer period of time abroad. The return of emigrated people has always been part of every migration movement – including recent times. Examples cover the return from overseas in the 19th and 20th century, the return of war veterans or the repatriation of war refugees during and after the First World War, or the return and repatriation of prisoners of war, refugees, exiles, concentration camp survivors, etc., during and after the Second World War.

Despite knowing about these return movements, migration was often viewed as a one-way process especially in academic studies, beginning with the 'uprooting' of people at the point of origin and ending with 'assimilation' into their adopted culture and country.[14] Still, recent research indicates that many

**12** http://anno.onb.ac.at/.

**13** Edda Currle (2006). "Theorieansätze zur Erklärung von Rückkehr und Remigration." In: *Sozialwissenschaftlicher Fachinformationsdienst soFid* Migration und ethnische Minderheiten 2006/2, pp. 7–23.

**14** Peterson Glen (2013). "Return migration." In: *The Encyclopedia of Global Human Migration*, pp. 1–5. DOI: 10.1002/9781444351071.wbeghm453.

people left their home countries with the notion of returning home at a certain point not so far in the future. This is also true for those who left their home voluntarily to resettle in other countries. People who emigrated by their own choice and later decided to come back to their homelands had different reasons for their decisions to return: success, failure, homesickness, rejection in the new country, changes in the economic or socio-political conditions in their native countries, or perhaps family members asked them to return.[15] For involuntary emigrants, some of these reasons may also apply, but their return was often supported, organized, or even forced by the country of origin. Although many voluntary and involuntary emigrants returned home disillusioned, with empty bags and ruined health, many of them also brought financial resources, new ideas, knowledge and skills. At the same time, those who had never left their native land often saw their return critically. The returnees were not always welcomed with open arms. There was mistrust of, for example, changed (political) ideas of return-migrants. Their return could increase problems on an already tight labour market, or be an additional burden on housing shortages. Those who left could also be seen as traitors who had abandoned those who had remained, or, just as well, as bearers of (lost) hope who brought promises of a better future with them.[16] Historical research on return migration, which is still rare, considers such patterns of, and motivations for, return migration,[17] examines historical, political, sociological or economical backgrounds,[18] or deals with autobiographical questions.[19]

So far, the complexity and heterogeneity of sources about return migration (e.g., letters, photos, reports, statistical material on return, snippets of personal

**15** Mark Wyman (2001). "Return migration – old story, new story." In: *Immigrants & Minorities*. Historical Studies in Ethnicity, Migration and Diaspora 20.1, pp. 1–18. DOI: 10.1080/02619288.2001.9975006.

**16** Annemarie Steidl, Wladimir Fischer-Nebmaier, and James W. Oberly (2017). *From a multiethnic empire to a nation of nations: Austro-Hungarian migrants in the US, 1870–1940*. Transatlantica (Innsbruck, Austria) volume 10. Innsbruck: StudienVerlag.

**17** Mark Wyman (2001). "Return migration – old story, new story." In: *Immigrants & Minorities*. Historical Studies in Ethnicity, Migration and Diaspora 20.1, pp. 1–18. DOI: 10.1080/02619288.2001.9975006.

**18** Kristina E. Poznan (2017). "Return Migration to Austria-Hungary from the United States in Homeland Economic and Ethnic Politics and International Diplomacy." In: *The Hungarian Historical Review* 6.3, pp. 647–667; Claudia Olivier (2013). "Brain Gain oder Brain Clash? Implizites transnationales Wissen im Kontext von Rückkehr-Migration." In: *Transnationales Wissen und Soziale Arbeit*. Ed. by Désirée Bender et al. Beltz Juventa, pp. 181–205; Marjory Harper (2012). *Emigrant homecomings: The return movement of emigrants*, 1600–2000. Studies in Imperialism MUP. Manchester: Manchester University Press.

**19** Katharina Prager and Wolfgang Straub, eds. (2017a). *Bilderbuch-Heimkehr? Remigration im Kontext*. Arco Wissenschaft Band 30. Wuppertal: Arco Verlag.

documentation, interviews, newspaper reporting) has complicated a structured analysis of the topic of return migration. The historical circumstances which brought a homecoming about were so diverse that generalizations are difficult to make and thus inherently faulty. What adds to the difficulty is the lack of empirical evidence. For example, it can still only be roughly estimated how many people who have emigrated voluntarily or involuntarily returned to their home countries.[20] The compilation of a corpus is therefore still an essential factor if the topic of re-migration is to be researched. This is not only true for global or transnational studies, but also for national and regional studies.

## 2.2 Discourse Analysis of Newspaper Reporting

Newspapers have an impact on the thinking, speaking and acting of a society.[21] For Norman Fairclough, newspapers are the 'predominant social field' for the creation of information, beliefs or arguments, which are necessary 'for establishing and sustaining economic, social and political systems and orders'.[22] Given the significance of news reporting for societies, it should come as no surprise that the discourses of newspaper reporting are essential objects of investigation.[23] There is no uniform definition of discourses or discourse analysis. Michel Foucault, who decisively shaped the concepts of discourse, defined a discourse as a 'group of statements that belong to a single system of formation'.[24] Based on this, humanities researchers have adapted the approach in various ways. For Michael Stubbs, analysing discourses means to 'study larger linguistic units such us conversational exchanges or written texts' whereby he defined discourse as 'language above sentences or above the clause'.[25] Jürgen Link described discourse as 'an institutionally consolidated concept of speech in as much as it determines and

**20** Katharina Prager and Wolfgang Straub (2017b). "Die Rückkehr zur Remigration. Zur Einleitung." In: *Bilderbuch-Heimkehr? Remigration im Kontext*. Ed. by Katharina Prager and Wolfgang Straub. Arco Wissenschaft Band 30. Wuppertal: Arco Verlag, pp. 9–19.

**21** Niklas Luhmann (1995). *Die Realität der Massenmedien*. Wiesbaden: VS Verlag für Sozialwissenschaften. DOI: 10.1007/978-3-663-16287-2.

**22** Norman Fairclough (2013). *Critical discourse analysis: the critical study of language*. Second Edition. New York: Routledge.

**23** John E. Richardson (2007). *Analysing Newspapers: An Approach from Critical Discourse Analysis*. New York: Macmillan International Higher Education.

**24** Michel Foucault (1969). *The Archaeology of Knowledge*. London: Routledge.

**25** Michael Stubbs (1983). *Discourse analysis: the sociolinguistic analysis of natural language*. Language in society 4. Chicago: University of Chicago Press; Oxford: Blackwell.

consolidates action and thus already exercises power',[26] and Ruth Wodak and Katharina Köhler, the front figures of critical discourse analysis, define discourse as social practices within a historical, political, socio-economic and cultural context.[27] (Critical) discourse analysis 'examines how texts represent and construct reality within a specific ideological system through implicit messages based on what is said and left unsaid'.[28] Discourse analysis therefore can help scholars to understand how social interaction shapes topics and concepts and how those change over time, whereas conceptual history, by comparison, focuses on the etymology and change of the meaning of words. Historical-semantic discourse analysis thus can open up semantic aspects and elements of knowledge that could escape a purely word-oriented history of meaning.[29]

## 2.3 Studying Discourses on Returnees in Newspapers and How to Get There

Reading newspapers, we can find different types of news coverage on return migration. The most frequent findings are reports on remigration and repatriation often written by journalists or repatriates. But also letters from returnees printed in newspapers can give interesting glimpses into return migration, as they provide a good insight into the thoughts of returnees. The following excerpt from the *Wiener Kurier* is part of a letter from a Jewish refugee talking about the fear of coming home to post-war Austria in 1945:

> Many may have gone away as Jews – but whoever comes back, comes as an Austrian! But this does not diminish the fear of those who return. Over seven years lie between us – and those experiences! Will it be possible to bridge this gap? Will we speak the same language? Will we not reproach each other, unconsciously?[30]

**26** Jürgen Link (1983). "Was ist und was bringt Diskurstaktik." In: *kultuRRevolution* 2, pp. 60–66.
**27** Ruth Wodak and Katharina Köhler (2010). "Wer oder was ist »fremd«?: Diskurshistorische Analyse fremdenfeindlicher Rhetorik in Österreich." In: *Sozialwissenschaftliche Studiengesellschaft* 2010.1, pp. 33–55.
**28** Susana de los Heros (2009). "Linguistic pluralism or prescriptivism? A CDA of language ideologies in Talento, Peru's official textbook for the first-year of high school." In: *Linguistics and Education* 20.2, pp. 172–199. DOI: 10.1016/j.linged.2009.01.007.
**29** Dietrich Busse (2008). "Begriffsgeschichte – Diskursgeschichte – Linguistische Epistemologie. Bemerkungen zu den theoretischen und methodischen Grundlagen einer Historischen Semantik in philosophischem Interesse anlässlich einer Philosophie der ‚Person'." In: *Diskurse der Personalität*. Ed. by Nikolaj Plotnikov and Alexander Haardt. Wilhelm Fink Verlag, pp. 115–142. DOI: 10.30965/9783846744321_009.
**30** "Wir sind quitt" (Oct. 13, 1945). In: *Wiener Kurier*, p. 4.

Next to letters, appeals and calls for help represent another type of coverage on return migration. Especially in the news coverage on returning prisoners of war, appeals for help were printed regularly in newspapers. Finally, a substantial part of the reporting on returnees consists of small advertisements. Here, returnees tried to sell land (often abroad), to find a partner for life, and asked for donations or a job. To describe oneself as a 'returnee' in these advertisements (Figure 1) seemed to be significant:


RÜCKWANDERER

verkauft 36 ha bestes Siedlungsland an Bahnstation,

BRASILIEN

Zahlung hier. Reichsbankgenehmigung erforderlich. Auskünfte unter „Parana 400" an das Ankündigungs-Büro dieses Blattes.

HEIRAT

Heimkehrer, Fünfzig., mit Beruf, wünscht Anschluß an nette gebildete Frau über 40 Jahre. Unter „Verlassen" an die Anz.-Annahme, Platzl 2.

8

**Fig. 1:** Small advertisements from returnees in Austrian newspapers from the 'Neue Freie Presse', 29 May 1938, p. 31 (left, 'Rückwanderer') and the 'Salzburger Blatt', 5 March 1946, p. 8 (right, 'Heirat'), retrieved from the ANNO platform (accessed in Dec 2020 at https://anno.onb.ac.at/).

Newspapers are therefore an interesting and available source to research return migration. They enable longitudinal studies, create a starting point for discourses on return migration, point to further primary sources and allow for an exploration of the discourses on how returning people were talked about in their country of origin: were they welcomed or perceived as a burden and threat? Can differences between groups of returning migrants be identified and were there changes over the time? How did returnees depict themselves?

Especially in reports, letters and appeals, we find several characteristic discourses on return migration, which include (implicit) arguments to support, promote, regulate or prohibit the return of people to their country of origin. Three main discourses can be identified: discourses on encouragement, authorisation or prohibition of return; discourses on motives of return like disappointment or deception; and discourses on benefits and threats of returnees for the country of origin.

Many examples of the discourse on encouragement of return can be found in 1907, when the governments of Austria-Hungary (and the Hungarian government in particular) became an active promoter of return migration to further an

economic and nationalist agenda.[31] The newspaper *Das Vaterland*, for example, printed the following paragraph in 1907:

> The Hungarian Farmers' Union, in association with the Hungarian Agricultural Association, has undertaken an action to promote the return of those who emigrated from Hungary to America and to facilitate their colonisation at home.[32]

While the return of emigrants from America was encouraged by the government, more critical voices called for control and prohibition of certain groups of returnees. In the following example from the *Salzburger Chronik für Stadt und Land* in 1907, it is suggested that only those people who were able to support themselves should be allowed to return:

> Through the leaders of the cooperatives, the trusted representatives in America are given the strictest instruction to encourage only those to return who have so much money in the domestic savings banks that they can acquire some land in their home country.[33]

Return migrants could help or endanger the government of the country of origin, both economically and politically. A return migrant might be someone who failed in the country he had emigrated to or someone who brought back skills, political ideas, or capital to invest in the homeland economy. Discourses about benefit or threat appear in many different historical and political contexts and usually come with specific (political or economical) intentions. Acceptance of returnees and a positive mood in the majority society was usually asked for because a political or economic benefit of this return was expected. On the other hand, there was rejection on the basis of assumed 'liberal' thoughts by those returning to Austria-Hungary, as the following excerpt from the *Salzburger Chronic für Stadt und Land* (November 1907) shows:

> In political terms, however, return migration will lead to a noticeable strengthening of the anti-Magyar nationalist movement, since the returnees understandably bring the American idea of freedom with them and carry it into the smallest towns and into the rural population.[34]

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**31** Kristina E. Poznan (2017). "Return Migration to Austria-Hungary from the United States in Homeland Economic and Ethnic Politics and International Diplomacy." In: *The Hungarian Historical Review* 6.3, pp. 647–667.

**32** "Kolonisierung von Auswanderern" (n.d.). In: *Das Vaterland* (), p. 11.

**33** "Ungarn-Die Rückwanderung aus Amerika" (Nov. 29, 1907). In: *Salzburger Chronik für Stadt und Land*, p. 2.

**34** "Die Rückwanderung aus Amerika" (Nov. 29, 1907). In: *Salzburger Chronik für Stadt und Land*, p. 2.

So, while digital methods can help us to find texts where return migration is talked about in one way or the other, they cannot (yet) label the discourses themselves: it is still the human eye that has to identify these discourses as being about encouragement or discouragement, motives, and advantages or disadvantages. This is 'partly because the way in which computers work is not automatically compatible with the way human brains work' as Hinke Piersma and Kees Ribbens phrased it.[35] As edmond noted, tools that focus on text do not reflect the practice of historians, which is based on broader information gathering and finding meaning in historical sources.[36] Nevertheless, digital methods can support discourse-analytical approaches,[37] especially when working with historical newspapers, because corpora containing discourses and arguments must be found before they can be identified, interpreted, and contextualized. Computational methods can strongly assist with the creation of adequate newspaper corpora for specific research questions and topics or when trying to find discourses. Without technical help, it is difficult to find, organise, and structure thousands of newspaper clippings. However, 'simple' procedures such as keyword searches, i.e. searching for specific articles using single or combined search terms, can be challenging because of the complex nature of language, changes in word meaning over time and polysemy. The following sections give an insight into the tricky challenges faced when creating a corpus on return migration using advanced keyword searches and frequency graphs.

# 3 The Keyword Challenge

## 3.1 Return Migration – Hard to Trace by Keywords

As emphasized in Section 2, discourse analyses of historical newspaper items depends on corpora that are at the basis of investigation. Interfaces can play a crucial role for the creation of corpora with search and collection building functions and easy-to-use download features. The process of collection building usually starts with complex search strategies designed to find relevant newspaper issues,

**35** Hinke Piersma and Kees Ribbens (2013). "Digital Historical Research: Context, Concepts and the Need for Reflection." In: *BMGN – Low Countries Historical Review* 128.4, pp. 78–102.

**36** Jennifer Edmond (2018). "How Scholars Read Now: When the Signal Is the Noise." In: *Digital Humanities Quarterly* 12.1.

**37** Andrew Ravenscroft and Colin Allen (2019). "Finding and Interpreting Arguments: An Important Challenge for Humanities Computing and Scholarly Practice." In: *Digital Humanities Quarterly* 13.4.

pages and articles given a research topic. Furthermore, due to the design and technology of interfaces, for historians working with digital newspapers, keywords are still key.[38] At the same time, keyword search goes hand in hand with some serious weaknesses for its use in historical research, which is caused by the complexity of language: alternative spellings, abbreviations, polysemy, changing word usage, idioms, misspellings or omissions of parts of words as well as sentences.[39] But this is only one part of the challenge, since for some topics we do not have simple keywords, as we will explain in the next section.

### 3.1.1 Concepts Do Not Equate to Single Words

Even when taking spelling variations and word changes into account, many search requests are difficult to define conceptually and hard if not impossible to trace by a simple keyword search alone. For the topic on return migration, there are a few specific German keywords that can be used in order to find relevant articles. These keywords mostly are variations of the word 'returnees', such as '*Heimkehrer*' [homecomers], '*Rückkehrer*' [returnees], '*Rückwanderer*' [return migrants], '*Heimgekehrten*' [repatriates].

If the (relative) frequency of these keywords is compared to the frequency of other migration-related keywords in ten Austrian newspapers, the result shown in Figure 2 (page 137) is obtained. The orange color shows the frequency of the variations of 'returnees', compared with variations of 'emigrants' (blue), 'refugees' (purple), and 'immigrants' (green).

The graph provides a rough overview of the evolution of the terms used to trace migration in Austrian newspapers between 1820 and 1920. These terms do not reflect the entire spectrum of reporting on migration, but they show the trickiness and complexity when using terms in connection with migration movements. Even if the terms used for return migration (orange line) seem to appear less frequently in the corpus than the words indicating emigration (blue line) or refugee movements (purple line), it is nevertheless not correct to deduce from the graph that return migration was less frequently reported on than the other issues. For example, many of the emigrants or refugees mentioned here are people returning to their communities of origin: They would be emigrants/refugees who return

**38** Eva Pfanzelter et al. (2021). "Digital interfaces of historical newspapers: opportunities, restrictions and recommendations." In: *Journal of Data Mining and Digital Humanities*. DOI: https://doi.org/10.46298/jdmdh.6121

**39** Sheila Bair and Sharon Carlson (2008). "Where Keywords Fail: Using Metadata to Facilitate Digital Humanities Scholarship." In: *Journal of Library Metadata* 8.3, pp. 249–262.

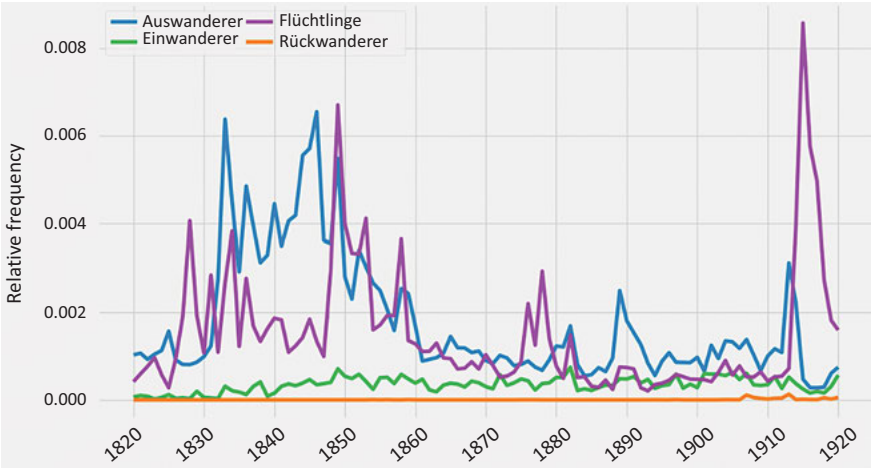



**Fig. 2:** Frequency graph of the terms *auswander** / *ausgewander** (*emigration/emigrated*), *einwander** / *eingewander** (*immigration/immigrated*), *flüchtling** / *geflüchtete** (*refugee/people who fled*), *rückwanderer** / *rückkehrer** / *heimkehrer** (*return migrant/returnee/homecomer*) in ten Austrian newspapers from 1820 to 1920.

home. The terms needed to identify the right group of people (the returnees) and/or the right discourses (those about return migration) would therefore be a combination of the words emigrants/immigrant/refugees and return/returning/coming home ('Auswanderer/Einwanderer/Flüchtlinge' and 'heimkommen/heimkehren/zurück kommen' and similar). This example – and there are many similar ones – shows that using single keywords or even bi-grams, if they were available in an interface search, is not enough when looking for re-migration issues.

### 3.1.2 Word Flexions

Combining keywords, as it is done when specific distances between words are taken into consideration (the so-called distance search) can help to find linguistic constructions such as 'refugees ... return home'. The ANNO interface enables users to find keyword combinations with defined word distances, a feature that is extremely helpful. However, word combinations do not always solve the problem. First, identifying the necessary terms to find the sought-for data is time-consuming and error-prone. In ANNO, this kind of search is even made more difficult because wildcards cannot be used in combination with the distance search and users are therefore obliged to try out many different word flexions and spelling

variants. For example, it is not possible to search for '*heimat *rückkehr**'~20 ('home *return*'~20), which would include different tenses of the verb 'to return' as well as the noun 'the return'.

### 3.1.3 Ambiguous Keywords

In addition, because of language ambiguity, many of the combined keywords, but also more ambiguous single keywords lead to results that have nothing to do with the original search request. For researchers eliminating these irrelevant results manually is not possible if a search request includes several thousands of hits. One of the ambiguous words for return migration is '*Rückwanderung*' ('return migration'), which can occur in very different contexts, as Figure 3 clearly shows:

| ‚Rückwanderung' relevant | ‚Rückwanderung' not relevant |
|---|---|
| ‚Allein in New=York drängen sich nun Tausende von Opfern der Krise, deren Mittellosigkeit ihnen die Rückwanderung bisher verwehrt hat, in die Vorzimmer der beteiligten Konsulate, um hier ihre Absicht zu erklären, „zu den Fahnen zu eilen", bei kostenfreier Ueberfahrt und zehn Heller Löhnung den Tag – soweit es sich um österreichisch= ungarische Reservisten handelt.' | 'Die mächtigen Dämme, die in den letzten Jahren vom Lido aus auf Kilometerlänge ins Meer hinausgebaut wurden, zwingen die Fluth zu rascherem, mächtigerem Eindringen in das Wirrsal der Lagunencanäte, und auf der Rückwanderung zum Urgewässer nimmt sie den aufgewühlten Schlamm mit sich und trägt ihn hinaus ins Freie.' |
| (In New York alone, thousands of victims of the crisis, whose pennilessness has so far prevented them from returning home, are now crowding into the antechambers of the consulates involved to declare their intention to 'rush to the flag', with free passage and then Heller a day – as far as Austrian-Hungarian reservists are concerned.) | (The massive dams that have been built in recent years from the Lido over a distance of several kilometers into the sea force the tide to penetrate the chaos of the lagoon channels more quickly and more powerfully, and on its way back to the primeval waters it takes the stirred up mud with it and carries it outside.) |

**Fig. 3:** Extracts from newspaper articles with the keyword '*Rückwanderung*', on the left an example of a relevant article and on the right a non-relevant article in connection with return migration.

Both the use of unique keywords and the combination of keywords make it difficult to create appropriate newspaper corpora. Unique keywords lead the researcher to only a restricted selection of articles, while the combination of keywords influences the search on the basis of the researcher's prior knowledge and oftentimes trigger irrelevant results.

Newspaper interfaces make language phenomena more apparent without actually creating them in the first place. Interfaces steer query and search processes and can therefore be seen as tools for source criticism but also as tools

that need to be reviewed and used critically. Interfaces can profit from some basic methods and tools, as the following sections show.

## 3.2 The Disappearance of the Second World War

One important first step when creating sub-corpora is to look at the frequency of words within a corpus. This can uncover the basic use and distribution of terms over time. They can give information on when a word or term was used and they can help to find entry points to corpora as well as to restrict the analysis to a specific period of time, if needed. With the help of diachronic frequency analyses, relative and absolute frequencies of pages, articles or words can be counted and thus information about the relevancy of texts over time or the distribution of certain linguistic patterns can be provided.[40] The ANNO platform offers no possibilities to create diachronic frequency graphs. The interface provides the number of pages where a keyword can be found and allows searching for specific years, however, the distribution of keywords over time is currently not traceable.

Still, even the amount of pages where a keyword appears can be helpful to gain a first impression of the search results. Using the absolute frequency of the keyword '*Heimat*' in combination with '*rückkehr/zurückkehren/heimkehren/zurückgekehrt*' ('home' combined with 'returning/return/return home/returned'), the graph in Figure 4 can be created:

Taking historical events into account, a closer look at the graph immediately reveals a discrepancy: The number of hits, not unexpectedly, rises sharply during and immediately after the First World War, roughly between 1915 and 1920. Very surprisingly, however, such a peak does not occur during the Second World War, between 1939 and 1945/46. This inconsistency and unlikely result also becomes apparent when using other search terms that should show a rising number of hits in the period after 1933. The question of why the results of the frequency analyses are distorted leads to a consideration and subsequent examination of the following questions:

- Were important search terms overlooked?
- Was there a change in the use of terms? Were new terms used after the beginning of the Second World War?

**40** Gregor Wiedemann, Matthias Lemke, and Andreas Niekler (2013). "Postdemokratie und Neoliberalismus – Zur Nutzung neoliberaler Argumentationen in der Bundesrepublik Deutschland 1949–2011." In: *ZPTh – Zeitschrift für Politische Theorie* 4.1.

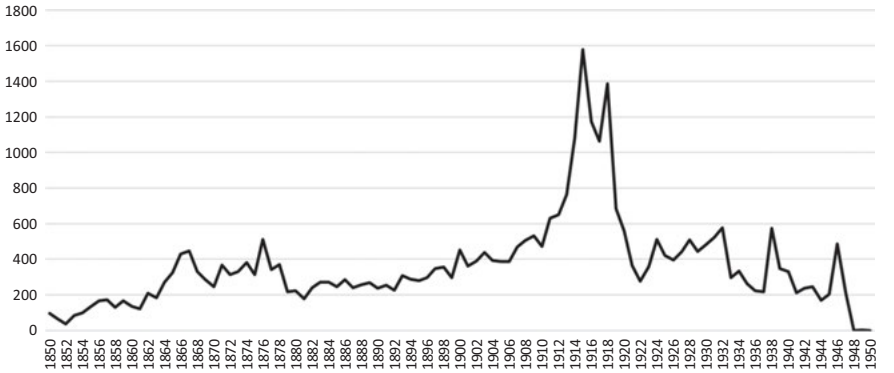



**Fig. 4:** Absolute frequency graph of '*Heimat*' ('home') in combination with '*rückkehr/zurückkehren/heimkehren/zurückgekehrt*' ('returning/return/return home/returned') (1850–1950) within a distance of 20 words. The distance search function in ANNO has been used to combine the terms.

- Is there a large discrepancy between the number of newspapers published during the First World War and those published during the Second World War?
- Is there a discrepancy between the number of found pages from the First World War and the Second World War that are available in digitised form?
- Is there a change in the quality of the digitised newspapers or the OCR for certain newspapers?

Careful browsing and comparison of two selected newspapers for this search and their OCR'd versions ultimately revealed that the OCR quality between 1938 and 1945 is extremely variable across newspapers and greatly differs from the one observed for time period 1915–1920. Figure 5 (page 141) shows an article on return migration in the newspaper *Neue Freie Presse* from 1938, with a page facsimile on the left and its barely decipherable OCR on the right. Apart from the headline, almost no combination of letters, numbers or signs is readable and many of the combined characters shown are not German words but seemingly random collections of letters and symbols.

## 3.3 Bringing Back the Second World War

Having worked with many newspaper interfaces, this outcome should not have been surprising. There have been ongoing discussions about the impact of the



**Rückkehr in die befreite Heimat**

Donnerstag mittag trafen in Reichenberg die ersten in die Heimat zurückgekehrten Flüchtlinge ein. Es handelt sich um 16 Männer aus dem Bereich der SA-Gruppe Mitte (Magdeburg), die während ihres Aufenthaltes unter dem Schutze des Altreiches von der Gruppe SA-gemäß ausgebildet, vollkommen eingekleidet und jetzt in ihre Heimat zurückgeleitet wurden. Zum Empfang versammelte sich die gesamte Bevölkerung auf dem Marktplatz, wo gegen Mittag sturmweise die Männer mit gepacktem Tornister, voran die Fahne, auf dem Adolf-Hitler-Platz aufmarschierten. Besonders stürmisch begrüßt von den Heimkehrern wie von den Reichenbergern wurde bei seinem Erscheinen der Reichskommissar. Nach der Begrüßung durch einen Vertreter der Stadt und nach einer Ansprache von SA-Obergruppenführer Kob, erklärte Konrad Henlein: Alle, die hier standen, seien bereit gewesen, ihr Leben für die Heimat in die Schanze zu schlagen. Daß sie nicht eingesetzt zu werden brauchten, dankten sie dem Führer. Konrad Henlein gedachte dann der Toten, die um Sudetendeutschlands Freiheit gefallen waren. Sein Dank galt den Betreuern der Flüchtlinge im Reich. Die Kameradschaft und begeisterte Aufnahme im Schutze des starken Reiches sollten die Männer durch ihre stete Einsatzbereitschaft danken. Nach dem Sieg-Heil auf den Führer zogen die Formationen unter Vorantritt des Musikkorps und des Spielmannzuges der Gruppe Mitte an Konrad Henlein und den Gruppenführern vorüber.

Rückkehr in die befreite Heimat
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**Fig. 5:** *Neue Freie Presse*, 15 October 1938 (fac-simile on the left, OCR text on the right).

OCR quality on research findings.[41] Still, the extent of the influence of poor OCR quality in combination with missing metadata (which will be discussed below) is striking. The following paragraphs will therefore introduce some methods, tools and functions that can help increase the reliability of the findings.

### 3.3.1 Data and OCR Quality

The OCR quality of the newspaper collections in ANNO not only changes significantly over time, but also varies considerably between different newspapers. Confronted with the findings above, researchers should become extremely cautious and skeptical about the results they get using digital newspaper interfaces and the functionalities implemented there. These irregularities lead to the frustrating conclusion that the full-text search results in ANNO – and in many other newspaper interfaces – are only part of what is in the data and that they are distorted by faulty OCR and missing data. With the OCR quality as poor as it is in the example above, not even good frequency analysis tools could help to

**41** Myriam C. Traub, Jacco van Ossenbruggen, and Lynda Hardman (2015). "Impact Analysis of OCR Quality on Research Tasks in Digital Archives." In: *Research and Advanced Technology for Digital Libraries*. Ed. by Sarantos Kapidakis, Cezary Mazurek, and Marcin Werla. Vol. 9316. Cham: Springer International Publishing, pp. 252–263.

create an interpretable picture. However, having a graph of relative frequencies next to the search results could point users to possible OCR flaws or similar problems in the dataset or the digitised material. This is important, because for researchers, faulty OCR does not necessarily have to be accepted as an unchangeable fact. Tools like Transkribus[42] can help to re-OCR documents and reach a text-quality that minimizes the impact of OCR on further analysis. Figure 6 on the right shows the improved, almost flawless OCR from the example shown in Figure 5 (*Neue Freie Presse* from 1938). The improved OCR was created with Transkribus using the downloaded, original scan from the ANNO platform. Interface providers can support this step by providing suitable download options (e.g., the possibility to download a whole newspaper issue).

| **OCR from the ANNO platform (ÖNB)**<br>(*Neue Freie Presse*, 15 October 1938) | **Improved OCR with Transkribus**<br>(*Neue Freie Presse*, 15 October 1938) |
|---|---|
| Rückkehr in die befreite Heimat<br>®onnerStag mittag trafen in IReidhcubcrg bie<br>e r fte n in bie .§cimat zurückgekehrten f5tkrhi^iitgo ein.<br>h®>^bclt sich 16 SJlänner auS bem Sereidt ber @91»<br>©nippe HUitte (ünagbeburg), bie währenb il)reS 9lufentholte§<br>unter bem ©d)uhe bcS 9lltreid)e§ oon ber ©ruppc '^»gemiä^<br>ausgebilbet, oollkommen eingekleibet unb je\|t in ihre .^eimat<br>Znrückgeleitet würben. ©mpfang oerfammelte sich bie<br>gesamte ©eoöllterung auf bem iDlarktplah, Wo gegen Sllittag<br>ftnrmweife bie SRäuner mit gepadrtem Tornister, Uoran bic<br>f^ahnc, auf bem 9tbolf»,§itler»ißlah aufmarfdperten. | Rückkehr in die befreite Heimat<br>Donnerstag mittag trafen in Reichenberg die<br>ersten in die Heimat zurückgekehrten Flüchtlinge ein.<br>Es handelt sich um 16 Männer aus dem Bereich der SA=<br>Gruppe Mitte (Magdeburg), die während ihres Aufenthaltes<br>unter dem Schutze des Altreiches von der Gruppe SA=gemäß<br>ausgebildet, vollkommen eingekleidet und jetzt in ihre Heimat<br>zurückgeleitet wurden. Zum Empfang versammelte sich die<br>gesamte Bevölkerung auf dem Marktplatz, wo gegen Mittag<br>sturmweise die Männer mit gepacktem Tornister, voran die<br>Fahne, auf dem Adolf=Hitler=Platz aufmarschierten. |

**Fig. 6:** *Neue Freie Presse*, 15 October 1938 (OCR retrieved from ANNO in December 2020 on the left, OCR by Transcribus on the right).

### 3.3.2 Relative and Absolute Frequency Graphs

One of the key features used in humanities research, as already mentioned above, is frequency analysis. Frequencies are used to gain insights into language use and to create balanced corpora for adequate analysis.[43] Absolute frequencies, as seen in Figure 4, can produce an unrealistic picture, as they only count the number of times a term appears in a corpus without putting the results in relation, e.g., with the total number of words, articles or pages for a certain time

**42** https://readcoop.eu/transkribus/.

**43** Jani Marjanen (2019). *What's the frequency, Kenneth?* URL: https://www.newseye.eu/ blog/news/what-s-the-frequency-kenneth/.

period or a certain newspaper. If many more newspapers were available during the First World War than during the Second World War, which is in fact the case in the ANNO platform, then absolute frequencies will always show a more frequent use of the term during the First World War, even if actually, in relation, many more articles were written about people returning to their country of origin during the Second World War.

Relative frequencies, on the other hand, can solve this problem by taking these relations into account. Using ANNO, relative frequencies cannot be generated automatically, as required metadata such as word counts or number of digitised pages/newspapers per year are not available. Metadata (e.g., pages per issue per year), however, is available at ONB Labs,[44] the Austrian National Library's digital laboratory. This makes a manually created frequency calculation (pages per year) possible, as can be seen in Figure 7:

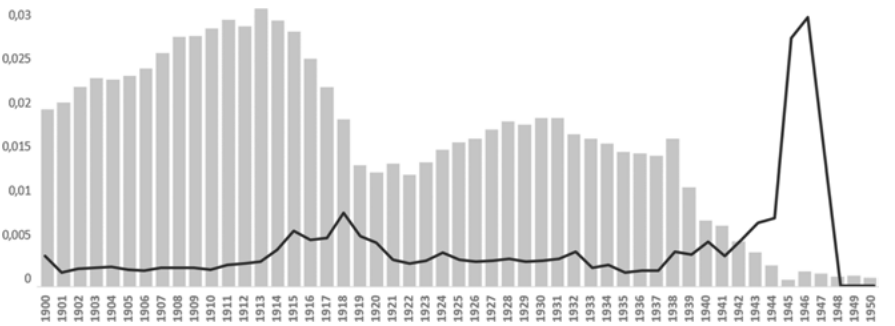



**Fig. 7:** Relative frequency graph (pages per year) of 'Home' combined with 'returning/return/return home/returned' (1900–1950).

As Figure 7 clearly shows, the number of hits obtained when researching remigration during the Second World War is significantly higher than the one during World War One. This outcome, which meets the expectations of historians, shows how important it is to critically question the information given on newspaper interfaces. It also underlines that such methods can be used to test and prove existing hypotheses. Metadata about digital collections provided on interfaces consulted by users is therefore very important. Missing metadata, as shown in Figure 4, as well as the lack of basic tools to analyse and evaluate findings affect

**44** *ONB Labs – Static Plotly Graphs* (n.d.). URL: https://labs.onb.ac.at/en/tool/sprachenin-anno/.

general faceting and browsing and therefore hinder the preparation of conclusive frequency analyses, which in turn could point users to some of the above-mentioned flaws.

### 3.3.3 Contextualization

Another important feature concerns contextualization of historical sources. Contextual information is essential for researchers in order to describe, explain, compare, or evaluate them. Interfaces could offer different kinds of context to users: Context on newspapers themselves, which sometimes is available already, can situate the document in time and space by including information on publication periods, publication places, editors, publishers, imprint details, political orientation as well as possible predecessor or successor newspapers. Frequency graphs (as can be seen in the example of return migration) can sometimes give context to the query and they can point to faulty data, keywords or dataset issues. Additional information on the newspapers and on datafication processes can help to understand and interpret the results better. Contextualization of the newspaper content, on the other hand, could be given through the recognition and linking of events or named entities (person, names, organizations or places) to external knowledge bases such as Wikidata. Automatic linking of newspaper content, however, can again create biases. Named entity recognition, for example, can create demographic biases (gender, ethnicity or race),[45] or biases caused by noisy text,[46] similarities in names or places, etc.

### 3.3.4 Transparency on Data and Tools

The call for perfectly clean data and appropriate tools should be preceded by the call for more transparency. This includes the traceability of:

- The search functions: are stemmed or exact words/word combinations being looked for and what does that imply? How does the search function work and what does it show?

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**45** Shubhanshu Mishra, Sijun He, and Luca Belli (2020). "Assessing Demographic Bias in Named Entity Recognition." In: *arXiv:2008.03415 [cs]*. URL: http://arxiv.org/abs/2008. 03415.

**46** Vinh-Nam Huynh, Ahmed Hamdi, and Antoine Doucet (2020). "When to Use OCR Post-correction for Named Entity Recognition?" In: *Digital Libraries at Times of Massive Societal Transition*. Ed. by Emi Ishita, Natalie Lee San Pang, and Lihong Zhou. Vol. 12504. Cham: Springer International Publishing, pp. 33–42. DOI: 10.1007/978-3-030-64452-9_3.

- The search results: what do the results show? Do hits represent the total number of matches or the number of pages/articles in which the matches occur and is there more than one match within one article?
- The data visualisations: are visualized results based on the total number of matches or pages/articles in which the matches occur? Are relative or absolute numbers visualized and how can they be interpreted?
- The text mining tools: what is the tool doing and what not? How are results being created and how can they be interpreted?

In order to give users a better command and a better understanding of the search process and search results, it would be an important feature to always give users the possibility to compare results with the original material. Again, transparency about datasets, search functionalities and data visualization could make gaps and empty spaces in the available collection visible and would enable the necessary source criticism.

### 3.3.5 Help Files, Information Pages and Best Practice Examples

In addition to the issues mentioned before, instructions, explanations and best practice examples are needed to guide users through the collections, the interfaces and digital tools. The better search functions, faceted searches, and use of interfaces are explained with different approaches, the better users will know how to interpret the results and how to use the interfaces to their advantage. We also found that creating educational material addressing the above-mentioned pitfalls and challenges for the libraries' use on the interfaces or in conjunction with their respective educational settings can help raise awareness to critical issues about data, interfaces and tools and attract new user groups.

## 4 The potential of Text Mining

While the above-mentioned methods seem to be fairly easy to be implemented in existing newspaper interfaces, there are several strong indicators that support calls for more elaborate user interfaces, tools and methods. While methods to achieve satisfactory search results including simple keyword searches and frequency graphs can help to get a general idea about a topic, more sophisticated techniques such as bi-gram searches or keywords in context can increase search outcomes significantly. Even simple word clouds chosen for specific time slices can point to changing

discourses within a researched topic. In addition, keyword suggestions based on word embeddings as provided by the *impresso* and *NewsEye* interfaces can help to find keywords that are semantically related to the users' query[47] and therefore open up new perspectives and possibilities.

As Figure 8 shows, the keyword suggestion function included in the beta version of the NewsEye platform, a project aiming at improving digital approaches to historical newspaper corpora, found many matches that improve the tedious manual search in ANNO and hinted at other issues that were not discovered previously, such as the combination of 'homecomers' ('*Heimkehrer*') with 'internees' ('*internierte*'), 'reservists' ('*reservisten*') and convalescent' ('*rekonvaleszent*'). All of these could indeed indicate discourses worth pursuing when working on issues of return migration.

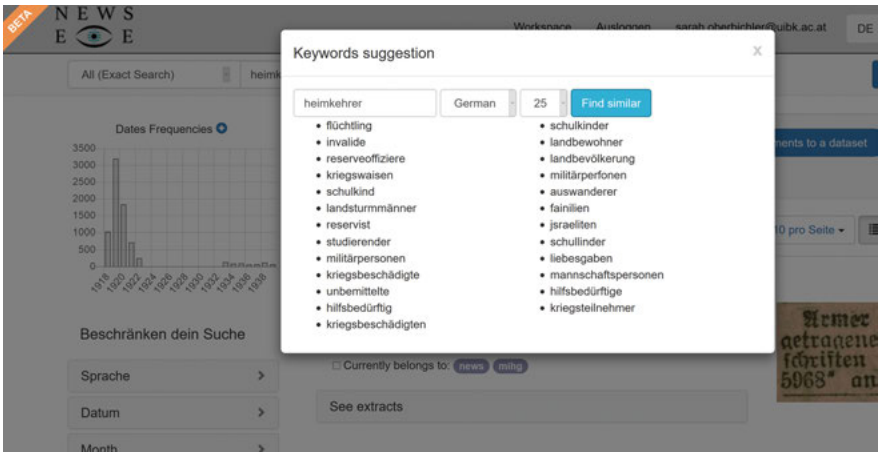



**Fig. 8:** Suggested keywords (set to 25) for the term 'homecomer' ('*heimkehrer*') produced by the NewsEye keyword suggestion tool as implemented in the beta version of the NewsEye platform (accessed in Dec 2020, https://platform.newseye.eu).

Also, text mining methods such as topic modeling, named entity recognition/linking or word embeddings can provide significant advantages when creating specific corpora and have the potential to become imperative for critical digital hermeneutics. For example, Federico Nanni, Simone Ponzetto and Laura Dietz used named entities and word embeddings to automatically build entity-centric

47 Saar Kuzi, Anna Shtok, and Oren Kurland (2016). *Query Expansion Using Word Embeddings*. Indianapolis Indiana USA.

event collections.[48] Their goal was to address the needs of humanities' and social sciences' scholars who work quantitatively on specific topics and events. With a similar aim in mind, researchers at the University of Helsinki are experimenting with approaches to adapting topic models and word embeddings on multilingual corpora.[49]

Text mining methods are also especially promising for the structuring, organisation and classification of text. For example, Pierre-Carl Langlais used a supervised topic modeling approach for genre classification, which allows the refining of searches to a specific genre.[50] Sarah Oberbichler, on the other hand, used methods that support article-based corpus building by taking the context (full content of the article) of keywords into account.[51] As discussed in Section 3, the corpus building for the topic of return migration is complicated by the fact that keywords are often either too narrow and leave part of the discourse in newspapers unseen, or too broad and thus lead to too many irrelevant results. Text classification can help group and subsequently filter articles that are actually relevant for the research question. This means that even ambiguous words can be used for the search without having to combine them with other terms, making the search less influenced by the researcher's prior knowledge. Using an approach combining Latent Topic Modeling (LDA) and the Jensen-Shannon Distance (JSD) method, articles about return migration and those on other, non-migration-related topics (as in Fig. 9) can be successfully and automatically grouped into and separated from each other.[52]

We do not argue that all text mining methods can or should be integrated in user interfaces, since it is difficult to find a 'one size fits all' approach for different user needs and specific research questions. However, interfaces that give access to huge historical datasets could greatly benefit from giving information on possibilities of further analysis options for the found material and by providing

**48** Federico Nanni, Simone Paolo Ponzetto, and Laura Dietz (2017). "Building Entity-Centric Event Collections." In: *2017 ACM/IEEE Joint Conference on Digital Libraries (JCDL)*. Toronto, ON, Canada, pp. 1–10.

**49** Elaine Zosa and Mark Granroth-Wilding (2019). "Multilingual Dynamic Topic Model." In: *International Conference on Recent Advances in Natural Language Processing ((RANLP 2019))*. URL: https://zenodo.org/record/4153232#.YyIQoN9CTy0.

**50** Pierre-Carl Langlais (2020). *Numapresse/TidySupervise*. URL: https://github.com/Numapresse/TidySupervise.

**51** Sarah Oberbichler (2020). *Using LDA and Jensen-Shannon Distance (JSD) to group similar newspaper articles*. URL: https://zenodo.org/record/3887193#.YyIvEt9CTy0.

**52** Sarah Oberbichler and Eva Pfanzelter (2021). "Topic-specific corpus building: A step towards a representative newspaper corpus on the topic of return migration using text mining methods". In: Journal of Digital History, 1.1. DOI: 10.1515/JDH-2021-1003.

| | Relevant_Text | 3 | Non_Relevant_Text | 0 |
|---|---|---|---|---|
| 0 | Ein Knirps unterm Tropenhelm\n Auf dem Anlegep... | 3 | „Ober wos denkst denn, so long ausbleibn!"\nFü... | 0 |
| 1 | Wenn trotzdem süber die Behandlung der Heimkeh... | 3 | Innsbruck, 31. Mai. Auf einer Pressefahrt der ... | 0 |
| 2 | Was die materialle Versorgung der Heimkehrer\n... | 3 | Nachdem Gauleiter Fritz Wächtler die Meldung d... | 0 |
| 3 | Die Austria wird diesmal mit einer starken Elf... | 3 | Der Stand der Sonne erinnerte Henriette daran,... | 0 |
| 4 | Walz war im Frühjahr 1920 von der öster\nreich... | 3 | Die äußeren Ursachen sind bekannt: Die Rückwan... | 0 |
| 5 | Das ist alles recht schön, aber jedenfalls\nei... | 3 | Millionen Soldaten, die im großen Weltkrieg fü... | 0 |
| 6 | Ein Fall auf dem Dampfer\nTriest, 6. Juni. Ges... | 3 | Repertoire des Deutschen Volkstheaters. Samsta... | 0 |
| 7 | Tomaschek war nach Kriegsausbruch eingerückt, ... | 3 | Seit Harald Paulsen das Theater am Nollendorfp... | 0 |
| 8 | Daß die Versorgung der Heimkehrenden in der\ne... | 3 | säule. Zugleich aber fiel das Barometer, und s... | 0 |
| 9 | Das gesamte deutsche Volk, vorab wir im Gau Ti... | 3 | .Vom 15. bis 22. Mai veranstaltete die Kreisle... | 0 |
| 10 | Rasch und doch sicher löste sie das Blatt\naus... | 0 | (Turistenunglück auf der Hohen Wand.)\nAus Wie... | 3 |

**Fig. 9:** Articles automatically grouped into relevant (*Relevant Text*) and non-relevant articles (*Non Relevant Text*) for the topic on return migration. For testing, articles were manually assigned to numbers (number 3 for relevant and number 0 for non-relevant articles) which proves whether the articles were selected correctly.

users with various export possibilities (.csv, .json, .xlsx) in order to allow them to apply advanced digital methods. Users, on the other hand, would greatly profit from such functionalities. It would sensitize them even more to faults and concerns, limitations and possibilities, and again, train them in source as well as tool criticism.

# 5 Conclusion

New digital tools and methods to support research, analysis and corpus building are currently being created, tested and evaluated for example by research projects such as '*impresso* – Media Monitoring of the Past'[53] or NewsEye[54]. Their aim is to enrich and connect historical newspapers of regional or national libraries with more advanced tools. Keyword suggestions, suggestions of related articles or topics, tools to build datasets and download options are powerful ways to support research with historical newspapers. The value of digitized newspapers for humanities increases significantly with these new developments. The project's interfaces take a first step towards collection building supported by machine

**53** https://impresso-project.ch/.
**54** https://www.newseye.eu/.

learning methods. They show how enriched data could ultimately improve access to historical newspaper collections. These interfaces, like some others, also aim at offering a variety of download options in order to facilitate further processing, e.g., the qualitative analysis of the created collections or further processing of the data outside the interface.

As we have tried to show, such efforts take digital hermeneutics seriously. A focus was put on pitfalls and challenges that come with corpus creation and thus with digital heuristics. To critically use, test and reflect the tools and outcomes remains an integral part of the research process. While individual users can often hardly understand how tools and methods work (and do not have to), they have to be given the information and tools to understand and critically question the outcomes. Therefore, digging into digital newspaper corpora continues to be an exciting field for interdisciplinary research. The contributions of humanities' critical approaches to the fields of computer science or digital cultural heritage curation are important, but their implications at this point cannot be grasped fully yet.

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